

Table 9: Performance comparison using manually created noisy documents.

Method	F1
Qwen2-VL-7B-SURf	64.07
Qwen2-VL-7B-DANCE	66.97

Table 10: Results on Fine-Grained Noise Benchmarks.

Method	MDC	CMC
SURf	65.5	58.2
DANCE	68.9 (+3.4)	59.5 (+1.3)

A Prompts

In this section, we provide the prompts used in this paper.

WebQA with Retrievals

You are a helpful question answerer who can provide an answer given a question and relevant context.

Question: [Question]

Context: [Context]

Provide a single sentence that answers the question based on the given context.

Answer:

WebQA without Retrievals

You are a helpful question answerer who can provide an answer given a question.

Question: [Question]

Provide a single sentence that answers the question.

Answer:

MultimodalQA with Retrievals

You are a helpful question answerer who can provide an answer given a question and relevant context.

Question: [Question]

Context: [Context]

Provide a single word or phrase that answers the question based on the given context.

Answer:

MultimodalQA without Retrievals

You are a helpful question answerer who can provide an answer given a question.

Question: [Question]

Provide a single word or phrase that answers the question.

Answer:

B Noise Rejection Ability

To verify the effectiveness of the noise rejection ability of our proposed method, we manually construct a validation set consisting of ground truth documents and irrelevant documents. Specifically,

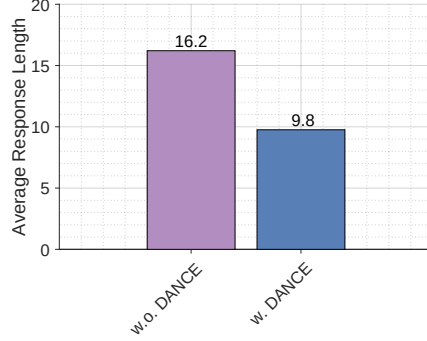


Figure 7: Comparison of average response lengths (in tokens) between models without DANCE instruction tuning (w.o. DANCE) and with DANCE instruction tuning (w. DANCE), demonstrating how instruction tuning leads to more concise responses aligned with downstream task requirements.

questions in WebQA have 1 or 2 ground truth documents. We randomly select 4 or 3 irrelevant to make sure each query has 5 documents. Then, we evaluate this dataset with SURf and our proposed method, as shown in Table 9. The results demonstrate that our method has the ability to ignore irrelevant documents in retrievals.

To create fine-grained noisy documents to verify the effectiveness, we create two fine-grained noisy benchmarks based on the MultimodalQA validation set, including cross-modal semantic conflicts (CMC) and contradictory information across multiple documents (MDC). For CMC, we prompt GPT-4o-mini to write the wrong text documents based on the ground truth captions of images. For MDC, we prompt GPT-4o-mini to identify entities or attributes in ground truth text documents and use wrong concepts to replace them. We get 521/472 samples with MDC/CMC, and the results on EM are presented in Table 10. The results show the robustness of our method regarding complex noise.

C Response Length

As shown in Figure 7, we observe a significant difference in average response lengths between models with and without DANCE instruction tuning. The *w.o. DANCE* model generates longer responses with an average length of 16.2 tokens, while *w. DANCE* model produces more concise responses, averaging 9.8 tokens. This reduction in response length can be attributed to the instruction tuning process aligning the model outputs with downstream task requirements. WebQA and MultimodalQA typically expect phrase-level or single-sentence responses, which are more concise than the text lengths used during pre-training or supervised fine-tuning stages. Through DANCE instruction tuning, MLLMs learn to generate responses that better match these dataset characteristics. The shorter response lengths contribute to improved inference efficiency, as MLLMs generate text in an autoregressive manner. With fewer tokens to generate, the models require fewer generation steps, resulting in reduced inference time. This efficiency gain, combined with Windsock’s adaptive retrieval strategy, helps optimize the overall computational performance of our system while maintaining response quality.

D Qualitative Results and Error Analysis

Figure 8 presents several examples from WebQA and MultimodalQA that illustrate both successful cases and typical failure modes of our system. Through analysis of these cases, we identify two main types of errors.

Misdirection from Windsock In some cases, Windsock fails to determine the optimal retrieval strategy. For example, in the “Ben Pizza” question, Windsock does not select the correct modality. However, DANCE-tuned MLLM can ignore the irrelevant contexts and generate a correct response.

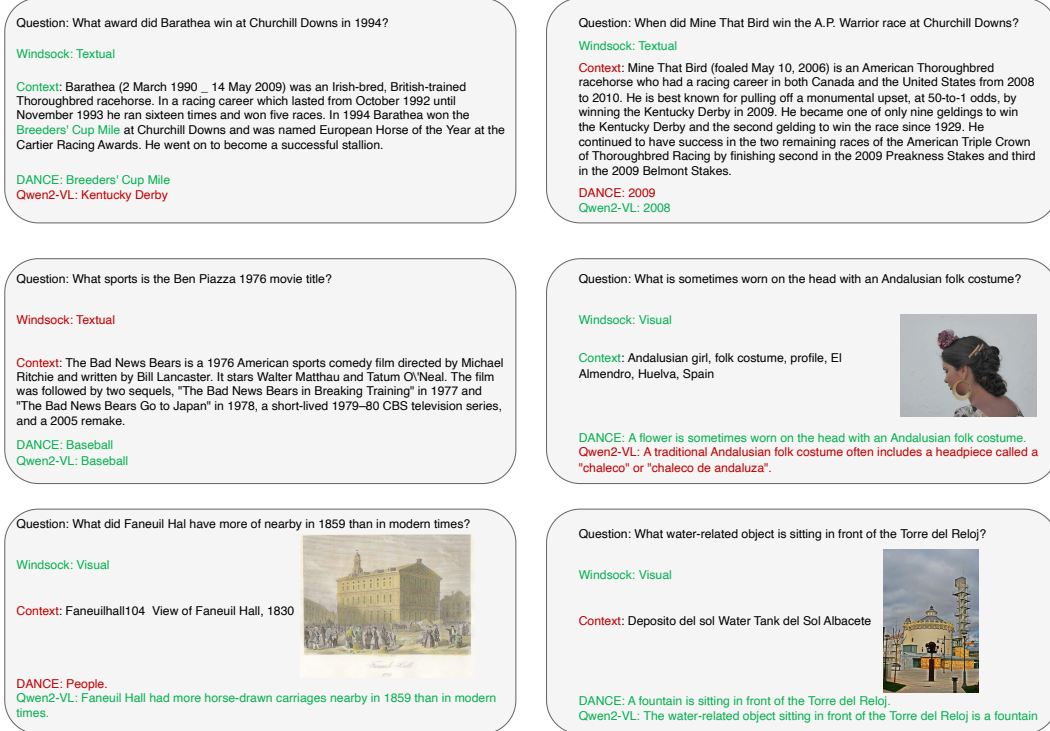


Figure 8: Qualitative examples from WebQA and MultimodalQA showcasing different response patterns between DANCE-tuned and base models. Each example includes the question, retrieved context, responses from both models, and Windsock’s modality selection. Examples illustrate both successful cases and common failure modes in our system.

Table 11: Performance comparison of different retrieval models.

Retriever	MRR@1	MRR@5	Recall@1	Recall@5	mAP@5	NDCG@5
CLIP ViT-L-14-336	9.72	13.69	8.18	17.14	-	-
VisualBGE Base	46.28	56.50	36.16	64.30	50.63	55.91
Marvel ANCE	55.44	65.72	43.59	73.95	59.20	64.67

Distraction from Generator Even when Windsock correctly identifies the retrieval strategy and relevant documents are retrieved, the generator can sometimes be misled by noisy information in the documents. For instance, in the "Faneuil Hal" question, the irrelevant retrieval misleads the DANCE-tuned MLLM.

E Different Retriever Performance Analysis

We evaluated three different retriever models to assess their effectiveness in our multimodal retrieval system: CLIP [20], VisualBGE [19], and Marvel [52]. The performance is measured using Mean Reciprocal Rank (MRR) and Recall metrics at two different cutoff points (top-1 and top-5). As shown in Table 11, Marvel demonstrates superior performance across all metrics, achieving the highest scores with an MRR@1 of 55.44% and Recall@5 of 73.95%. This indicates its strong ability to rank relevant documents highly and retrieve them effectively. VisualBGE shows competitive performance as the second-best option, with an MRR@1 of 46.28% and Recall@5 of 64.30%. In contrast, CLIP ViT-L-14-336 exhibits significantly lower performance across all metrics, suggesting it may be less suitable for this specific retrieval task. In this paper, we use VisualBGE as the default retriever, other retrievers [53, 54, 55, 56] may increase the system’s performance and it can be future work.

Table 12: More results using different backbones of Windsock.

Model	F1
DistilBERT-Base	55.5
CLIP ViT-B/16	49.2
Flan-T5-Small	58.3 (+2.8)

Table 13: Hybrid Retrieval Results.

Retrieval	F1	Ratio
NA	56.8	10.3
Visual	58.0	28.5
Textual	56.7	36.3
Hybrid	57.1	24.9
Windsock	59.1 (+1.1)	-

F More Results of Using Different Backbones of Windsock

Windsock is a novel query-dependent module in which the input is the query, and the outputs are the decisions on retrieval necessity (when) and modality selection (what). Compared to previous fixed and modality-agnostic methods, Windsock effectively reduces computational overhead and improves response quality. In this paper, we use Flan-T5-Small as the default model. We conduct experiments using DistilBERT and CLIP. The results presented in Table 12 indicate that Flan-T5-Small outperforms DistilBERT and CLIP since DistilBERT is a smaller model than Flan-T5, and CLIP is trained for the image/text alignment task. We believe the understanding ability of backbone models is a key point to improve Windsock’s performance. A larger model could perform better, but it also introduces more inference latency.

G Hybrid Retrieval and More Modalities

Existing dataset construction pipelines, like WebQA [39] and MultimodalQA [40], typically instruct annotators to create questions answerable by either images or text rather than requiring both simultaneously. Encyclopedic-VQA [41] and InfoSeek [42] require models to identify entities in images and retrieve knowledge from Wikipedia to get textual descriptions. Answering questions in these datasets does not need to retrieve both visual and textual information. It is a limitation of dataset construction, not a limitation of Windsock. We notice a benchmark, MRAMG-Bench [57], requiring models to always retrieve multimodal information for multimodal generation. Using WebQA’s visual retrieval, WebQA’s textual retrieval, and MRAMG-Bench, we randomly sample 2,000 data points from each as the training set. Then, we randomly sample 300 data points from each as the validation set. Extending Equation (2), we train a Windsock supporting NA, Visual, Textual, and Hybrid retrieval, as shown in Table 13. These results demonstrate the scalability of Windsock while maintaining the superior inference time.

The scalability of Windsock makes it easy to expand to more modalities. We use image and text modalities because they are the most common in MLLMs. We survey the RAG datasets and do not find one that supports more than three modalities. Inspired by the video classification task, we observe that WebQA contains questions that require more than 2 images. We concatenate these images and view them as a “pseudo video” to create the third modality in WebQA. We randomly sample 2,000/300 data points from each modality as the training/val sets. To build the video indexing, we use VBGE to extract image features and add them as the video embedding. The results are presented in Table 14. These results shows that Windsock’s flexibility of modalities and can be scaled up on more modalities.

Table 14: Results using More Modalities.

Retrieval	F1	Ratio
NA	47.1	9.1
Visual	49.8	29.4
Textual	45.5	46.8
Video	49.4	14.7
Windsock	51.0 (+1.2)	-

Table 15: Performance Comparison of using GPT-4o and Windsock as the decision module.

Method	EM
GPT-4o	40.7
Windsock	43.2 (+2.5)

Table 16: Results on Enc-VQA using ReflectiVA and the optimal Windsock.

Method	E-VQA	NA Ratio
ReflectiVA	35.5	0%
+ Windsock	36.9 (+1.4)	7.6%

H Robustness of Self-Assessment

Our method improves the robustness while maintaining efficiency in two complementary ways: For cases where retrieval transforms incorrect answers to correct ones, Windsock learns to identify these patterns and trigger appropriate retrieval strategy, addressing scenarios where the MLLM’s parametric knowledge is insufficient. For cases where answers remain incorrect despite retrieval since we comprehensively evaluate the response with all retrieval scenarios in Equation (3) and Equation (5), it indicates that it is too difficult to answer for the MLLM, which requires larger MLLMs or continual pre-training techniques to improve the performance of base models. We assign “NA” labels to skip retrieval to improve efficiency. Windsock ensures our system learns from successful retrieval patterns while making practical decisions for challenging cases that represent current model limitations. Moreover, this self-assessment method successfully connects the internal parametric knowledge with the downstream dataset preference through Windsock. We randomly sample 4,000/600 training/val samples from MultimodalQA. Instead of Windsock, we prompt GPT-4o to make the retrieval decision with a question. The results are presented in Table 15.

We analyze the errors of GPT-4o and find two error patterns. First, GPT-4o is more powerful than the generators in this paper, e.g., Qwen2-VL-7B. Questions that require retrievals for Qwen2-VL may be directly answered by GPT, leading to a wrong NA label. Second, some questions can be answered by either images or texts from GPT’s perspective. Due to the dataset bias, one of the modalities may not exist in downstream datasets, leading to wrong modality selection. Self-assessment and Windsock capture the bias and successfully make correct decisions regarding retrievals by connecting the internal parametric knowledge with the downstream dataset preference.

I Compared to ReflectiVA

ReflectiVA [14] retrieves entity textual descriptions that do not involve scenarios that require modality selection. It uses the MLLM as a point-wise reranker, first using the MLLM to decide if the retrieval is triggered and then generating a special token to filter out irrelevant documents one by one, significantly increasing the inference latency, resulting in an overall inference time of 5 to 7 times slower than our method (~ 80 GPU hours using H100). ReflectiVA achieves nearly 0% NA retrieval because it simply assigns Retrieval/NA labels to training samples in Infoseek, E-VQA, and LLAVA datasets. This fixed and modality-agnostic method cannot handle the increasing MLLM’s abilities compared to our method, which connects the internal parametric knowledge with the downstream dataset preference. Notably, using ReflectiVA as the generator, Windsock is compatible with ReflectiVA, as shown in Table 16.

In the main experiments, We use WebQA and MultimodalQA since they contain samples that require image or text retrieval, which is a more challenging and realistic scenario. INFOSEEK [42] and Enc-VQA [41] require models to identify entities in images and retrieve knowledge from Wikipedia to get

Table 17: Results on the subset of Enc-VQA.

Retrieval	Performance
NA	17.0
Textual	19.6
Windsock	20.6 (+1.0)

Table 18: Mixing general instructions improves the general ability.

MLLM	MME-P	MME-C	MMMU	MMStar
Qwen2-VL-7B	1423.25	484.00	54.1	60.7
+ DANCE	1394.05	435.71	49.7	56.1
+ DANCE + LLaVA-Instruct	1401.39	457.91	51.3	57.3

Table 19: Statistics on the knowledge bases.

Dataset	# Visual	# Textual
WebQA	403,277	544,489
MultimodalQA	57,052	218,258

textual descriptions, which do not involve scenarios that require modality selection. Windsock can be used to decide when to retrieve these datasets. To verify the effectiveness and efficiency of Windsock on them, we randomly sample from Enc-VQA and obtain 2,000/200 samples as training/val sets to train the Windsock. The generator is pre-trained LLaMA-3.1-8B. The results are presented in Table 17. The results demonstrate the effectiveness and efficiency of Windsock on various datasets.

J General Ability Improvement

There is a performance trade-off between specific tasks and general tasks: tuning on downstream tasks may lead to decreased performance on general benchmarks, as demonstrated by recent works. To improve the RAG performance while maintaining the general ability, we mix the instructions from Equation (5) with the LLaVA-Instruct dataset and fine-tune the model. We also add two extra general benchmarks MMMU [58] and MMStar [59]. The results are shown in Table 18. This demonstrates that mixing general instruction data could improve the instruct-tuned models’ general ability.

K Knowledge Base Construction and Windsock Decision

The visual knowledge base contains image file paths and corresponding captions. The textual knowledge base contains text chunks. We use IndexFlatL2 from faiss library to construct the indexing. The number of data in databases is listed in Table 19.

In our implementation, we use VBGE-base as our default multimodal retriever, which supports both modalities with a unified embedding space. During the preprocessing (indexing) stage, we use VBGE to encode images from the visual database and texts from the textual database to embedding vectors and store them in the corresponding vector database. During the retrieval stage, the query text with an optional query image is encoded into the query embedding vector. If Windsock selects visual/textual retrieval, we calculate the cosine similarity between the query vector and embedding vectors from the visual/textual database and return top k results. We will add these implementation details to the revised manuscript, including pseudo code for the retrieval process.

L Limitations and Future Work

While our proposed method shows promising results, several limitations should be noted. First, in this work, we only focus on the vision and language modalities. Our method can extend to more modalities like audio or tabular data, which can be used in future work to verify the effectiveness and efficiency of more modalities. Second, we do not elaborate on a complex pipeline to solve the single-hop and multi-hop questions separately. Specific designs for multi-hop questions may further improve the performance. Third, Windsock currently makes binary decisions about modality selection (visual vs. textual). This may be suboptimal for queries that could benefit from varying degrees of multimodal information. Future work could explore more nuanced retrieval strategies that allow for weighted combinations of different modalities.

M Broader Impacts

The Windsock framework introduced in this paper has several significant positive societal and technological impacts. This approach substantially reduces computational overhead while improving response quality by intelligently determining when retrieval is necessary and which modality to select. This efficiency gain reduces energy consumption and carbon footprint, supporting more sustainable AI deployment.

The adaptive nature of Windsock makes advanced multimodal AI systems more accessible to users with limited computational resources, helping democratize access to state-of-the-art AI technologies. In educational settings, improved response accuracy with reduced latency enhances the learning experience, while in research environments, it enables more efficient information retrieval and knowledge synthesis across modalities.

Furthermore, the DANCE instruction tuning approach improves robustness against misinformation by teaching models to better distinguish between relevant and irrelevant retrieved information. This framework’s modular design allows seamless integration into various existing systems, maximizing its practical impact across different applications and domains.

However, the system may inadvertently amplify existing biases in training data through its selective retrieval decisions, potentially reinforcing problematic information patterns. Additionally, as the technology becomes more widely adopted, there is a risk of increasing societal dependency on automated information filtering systems that operate as black boxes to most users.